

Generation Interconnection System Impact Study Report

For Queue Project Q21-1929

Nacoochee 138 kV

June 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1929. The Interconnection Customer proposes a 429.0 MW natural gas combined cycle facility interconnecting at the Nacoochee 138 kV substation in the Virginia area. The request was received on November 09, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 539 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,532,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: S TEXA / NACOOCHEE 138kV

Contingency: PARKIN / OPOSSUM CRK 138kV

Loading: 124.6% (Rating: 659 MVA)

DFAX: 57.7%

Pre-project loading: 87.0%

2. Facility: LA JOIE / MONEL 138kV L

Contingency: FRANKLIN-ELK GROVE SUBSTATION / NEW LONDON SUBSTATION 138kV

Loading: 102.2% (Rating: 212 MVA)

DFAX: 11.6%

Pre-project loading: 78.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STEXA / NACOOCHEE 138kV	Rebuild line with higher-capacity conductor	\$5,232,000	33% (\$1,713,000)
MONEL / LA JOIE 138kV	Reconductor line	\$7,954,000	23% (\$1,819,000)
	Total allocated to project		\$3,532,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.